AI ASSISTED CODING

ASSIGNMENT – 18.1

NAME: Aravind Reddy

HTNO: 2303A51027

Task 1 – Weather API Integration Connect to a Weather API to fetch current weather details for a given city. Instructions: • Prompt AI to generate Python code using the requests library. • Handle errors: invalid city name, API key errors, and network failures. • Display temperature, humidity, and weather description in a readable format. Expected Output: • A working Python script that fetches and displays weather data with proper error handling.

```python
import requests

def get_weather_description(code):
    descriptions = {
        0: "Clear sky",
        1: "Mainly clear",
        2: "Partly cloudy",
        3: "Overcast",
        45: "Fog",
        48: "Depositing rime fog",
        51: "Light drizzle",
        53: "Moderate drizzle",
        55: "Dense drizzle",
        56: "Light freezing drizzle",
        57: "Dense freezing drizzle",
        61: "Slight rain",
        63: "Moderate rain",
        65: "Heavy rain",
        66: "Light freezing rain",
        67: "Heavy freezing rain",
        71: "Slight snow fall",
        73: "Moderate snow fall",
        75: "Heavy snow fall",
        77: "Snow grains",
        80: "Slight rain showers",
        81: "Moderate rain showers",
        82: "Violent rain showers",
        85: "Slight snow showers",
        86: "Heavy snow showers",
        95: "Thunderstorm",
```

```python
    return descriptions.get(code, "Unknown")
def get_coordinates(city):
    url = f"https://geocoding-api.open-meteo.com/v1/search?name={city}"
    response = requests.get(url)
    data = response.json()
    if "results" in data and len(data["results"]) > 0:
        return data["results"][0]["latitude"], data["results"][0]["longitude"]
    else:
        raise ValueError("City not found")
def get_weather(latitude, longitude):
    url = f"https://api.open-meteo.com/v1/forecast?latitude={latitude}&longitude={longitude}&cur
    response = requests.get(url)
    data = response.json()
    if "current" in data:
        current = data["current"]
        weather_code = current["weather_code"]
        description = get_weather_description(weather_code)
        return {
            "temperature": current["temperature_2m"],
            "humidity": current["relative_humidity_2m"],
            "description": description
        }
    else:
        raise ValueError("Weather data not found")
if __name__ == "__main__":
    city = input("Enter a city name: ")
    try:
        latitude, longitude = get_coordinates(city)
        weather = get_weather(latitude, longitude)
        print(f"Current weather in {city}:")
```

```python
if __name__ == "__main__":
    city = input("Enter a city name: ")
    try:
        latitude, longitude = get_coordinates(city)
        weather = get_weather(latitude, longitude)
        print(f"Current weather in {city}:")
        print(f"Temperature: {weather['temperature']}°C")
        print(f"Humidity: {weather['humidity']}%")
        print(f"Description: {weather['description']}")
    except ValueError as e:
        print(e)
```

PROBLEMS OUTPUT TERMINAL PORTS

TERMINAL

```
...
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC> & C:/Users/varsh/AppData
rd/AIAC/lab_assignment_18.py
Enter a city name: Hanamkonda
Current weather in Hanamkonda:
Temperature: 35.6°C
Humidity: 38%
Description: Clear sky
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC>
```

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API downtime.
- Show conversion results clearly in tabular format.

Expected Output:

- A script that converts INR to multiple currencies with graceful error handling.

```python
import requests
def convert_currency(amount, from_currency, to_currency):
    url = f"https://api.exchangerate-api.com/v4/latest/{from_currency}"
    response = requests.get(url)
    data = response.json()
    if "rates" in data and to_currency in data["rates"]:
        rate = data["rates"][to_currency]
        return amount * rate
    else:
        raise ValueError("Currency not found")
if __name__ == "__main__":
    amount = float(input("Enter the amount in INR to convert: "))
    to_currency = input("Enter the currency to convert to (USD, EUR, GBP): ").upper()
    try:
        converted_amount = convert_currency(amount, "INR", to_currency)
        print(f"{amount} INR is equal to {converted_amount:.2f} {to_currency}.")
    except ValueError as e:
        print(e)
```

PROBLEMS OUTPUT TERMINAL PORTS

```
TERMINAL
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC> & C:/Users/varsh/AppData/Local/Programs/Pytho
rd/AIAC/lab_assignment_18.py
Enter a city name: Hanamkonda …
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC> & C:/Users/varsh/AppData/Local/Programs/Pytho
rd/AIAC/lab_assignment_18.py
Enter the amount in INR to convert: 50
Enter the currency to convert to (USD, EUR, GBP): USD
50.0 INR is equal to 0.53 USD.
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC>
```

Conversion results:

Currency	Rate (INR->Target)	Converted Amount
USD	0.012345	1.23
EUR	0.011234	1.12
GBP	0.009876	0.99

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars,forks, issues).

Instructions:• Prompt AI to write a Python function to send GET requests to GitHub API.

• Handle rate limit errors, 404 (repo not found), and invalid input.• Display repository name, description, stars, forks, and open issues.Expected Output: A Python program that retrieves and displays GitHub repositoryinformation with robust error handling.

```python
import requests
def get_github_repo_details(owner, repo):
    url = f"https://api.github.com/repos/{owner}/{repo}"
    response = requests.get(url)
    if response.status_code == 200:
        data = response.json()
        return {
            "name": data.get("name", "N/A"),
            "description": data.get("description", "N/A"),
            "stars": data.get("stargazers_count", 0),
            "forks": data.get("forks_count", 0),
            "open_issues": data.get("open_issues_count", 0)}
    else:
        raise ValueError("Repository not found")
if __name__ == "__main__":
    owner = input("Enter the GitHub repository owner: ")
    repo = input("Enter the GitHub repository name: ")
    try:
        details = get_github_repo_details(owner, repo)
        print(f"Repository Name: {details['name']}")
        print(f"Description: {details['description']}")
        print(f"Stars: {details['stars']}")
        print(f"Forks: {details['forks']}")
        print(f"Open Issues: {details['open_issues']}")
    except ValueError as e:
        print(e)
```

PROBLEMS OUTPUT TERMINAL PORTS

TERMINAL

```
Enter the GitHub repository name: AIAC_2303A51103
Repository Name: AIAC_2303A51103
Description: None
Stars: 0
Forks: 0
Open Issues: 0
```

Task 4 – Real-Time Application: News Headlines Aggregator

Scenario: Build a News Aggregator using a free News API.

Requirements:• Fetch top 5 headlines for a given category (sports, technology, health).• Handle errors like invalid category, missing API key, and request failures.• Display headlines in a user-friendly numbered list format.• Include a retry mechanism if the first request fails.Expected Output:• A working Python script that retrieves and displays news headlines with proper error handling

```python
import requests
NEWS_API_KEY = "YOUR_API_KEY"  # Replace with your own key from https://newsapi.org/
SAMPLE_HEADLINES = [
    {"title": "Sample: Local team wins championship"},
    {"title": "Sample: New tech product launches"},
    {"title": "Sample: Health officials advise daily walking"},
    {"title": "Sample: Stock market closes higher"},
    {"title": "Sample: Global conference schedules released"}
]
def get_top_headlines(category):
    if NEWS_API_KEY == "YOUR_API_KEY":
        print("WARNING: No NewsAPI key configured. Returning sample headlines.")
        return SAMPLE_HEADLINES
    url = f"https://newsapi.org/v2/top-headlines?category={category}&apiKey={NEWS_API_KEY}&pageSize=5"
    try:
        response = requests.get(url, timeout=10)
        response.raise_for_status()
        data = response.json()
        if data.get("status") != "ok":
            raise ValueError(f"NewsAPI error: {data.get('message', 'unknown error')}")
        if "articles" in data and data["articles"]:
            return data["articles"][:5]
        else:
            raise ValueError("No news found for the given category")
    except requests.exceptions.RequestException as e:
        print(f"Request failed: {e}")
        return None
    except ValueError as e:
        print(f"Error: {e}")
        return None
def print_headlines_table(headlines, category):
    if not headlines:
        print("No headlines available.")
        return
    print(f"\nTop 5 headlines in {category.capitalize()}:\n")
    print("Index | Headline")
```

```python
    print("------|---------")
    for i, article in enumerate(headlines, start=1):
        title = article.get("title", "N/A")
        print(f"{i:>5} | {title}")
if __name__ == "__main__":
    category = input("Enter a news category (sports, technology, health): ").lower().strip()
    if category not in {"sports", "technology", "health"}:
        print("Invalid category - defaulting to 'technology'.")
        category = "technology"
    headlines = get_top_headlines(category)
    if headlines is None:
        print("Retrying...")
        headlines = get_top_headlines(category)
    print_headlines_table(headlines, category)
```

PROBLEMS OUTPUT TERMINAL PORTS

TERMINAL

```
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC> & C:/Users/varsh/AppData/Local/Programs/Python/Python31
Failed to fetch news headlines.
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC> & C:/Users/varsh/AppData/Local/Programs/Python/Python31
arsh/OneDrive/ドキュメント/third/AIAC/lab_assignment_18.py
Enter a news category (sports, technology, health): sports
WARNING: No NewsAPI key configured. Returning sample headlines.

Top 5 headlines in Sports:

Index | Headline
------|---------
    1 | Sample: Local team wins championship
    2 | Sample: New tech product launches
    3 | Sample: Health officials advise daily walking
    4 | Sample: Stock market closes higher
    5 | Sample: Global conference schedules released
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC>
```